CO1-ATI - Descriptive Exam

CSA3714 - Software Testing

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4. Defect clustering:

It is a software testing principle that states that most defects are usually found in a small number of modules. Identifying these modules helps testers focus their efforts effectively.

Applications:

→ Testing shows the Presence of Defects:
Testing helps identify defects but cannot prove that software is completely defect free

→ Early Testing: Start testing during the requirement and design phases to detect defects early

→ Pesticide Paradox: Regularly update test cases to identify new type of defects.

Test Case Prioritization: Test case should be prioritized based on risk, business importance, frequency of use, and previous defect history.

Real-Time Example: In a online banking application, most defects may occur in fund transfers. Testers should prioritize fund transfer payment validation and transaction security before testing less critical features.

5. Defect Classification and Repository Manage

A defect repository is a centralized storage system used to record, track and analyze software defects. Effective repository management helps prevent recurring defec

Classification of Defects:

1. Functional Defects: Features do not work as expected
2. Performance Defects: The application respond slowly
3. Security Defects: The application contain security weakness.
4. Interface Defects: Communication betwee modules fails
5. Usability Defects: The application is difficult to use

Improvements to Prevent Recurring Defects:

→ Maintain a centralized defect repository

→ classify defects by type, severity, module and root cause

→ share defect reports with developers and testers.

11. Data Inconsistency in a Healthcare Application.

So, the data Inconsistency commonly occur in the Healthcare. It can affect patient records, prescriptions and medical reports

Defects may originate from incorrect database queries, coding errors, improper data synchronization.

Systematic Testing for Data Integrity:

1. Unit Testing: Check individual modules for data processing errors.
2. Integration Testing: Verify that defects before it actually errors that can able
3. Database Testing: Verify that data is stored, updated and retrieved.
4. Regression Testing: Ensure that fixing one defect does not introduce new defects

Real-Time Ex: A patient's blood group is updated in the patient registration module but remains unchanged in the laboratory module Integration testing and database validation can identify the synchronization problem.

12. Defect Repository to analyze issue

A defect repository stores information about software defects found during testing and after release. If a company does not analyze this information the same defects may occur repeatedly.

Identifying Defect Patterns:

* Modules with the highest number of defects
* Defects that occur repeatedly across releases
* Common coding and design mistakes
* Defects introduced after software changes.

Improvements to Prevent Recurring Defects:

→ Maintain an updated defect repository.
→ Perform root cause analyses for recurring defects
→ Create regression tests for previously fixed defects.
→ Automate frequently repeated tests.

Real time example In banking login failures occurs after every software update. By analyzing the defect repository the team identifies changes to authentication code to verify.

Web based Learning System Failure During Peak Usage

A web-based learning system may fail during peak usage when many students access videos, submit assignments or attend online examinations at the same time.

Gaps in Testing:

The team may have focused only on functional testing without checking system performance under heavy traffic.

Importance of Test Planning & Coverage.

* Test Planning: Define testing objectives, resources, timelines, and expected user loads.
* Load Testing: Verify system performance under expected peak traffic
* Performance Testing: Measure response time, throughput and resource usage.

Real-Time Ex: During an online examination, 5000 students attempt to log in it causing the website to crash. Load testing before the examination could identify servers capacity and improve the quality.